

Dexter Fernandes

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Backend Software Engineer

Experienced Backend Software Engineer with 3.5 years of end-to-end Python and C++ development, collaborating with cross-functional teams to ship edge computer-vision solutions for clients like British Petroleum and Austin Industries. I own the full ML lifecycle, i.e. managing large datasets with DVC, building PyTorch models, tracking experiments with MLflow, and packaging and deploying models via Azure shared file systems. I have refactored monoliths into Kafka-driven microservices that ingest 92 sensor values per second for time-series classification with sub-second inference, quantised for multi-system rig-floor safety management boosting detection accuracy by 15% and improving MTBF by 35%.

WORK EXPERIENCE

Esbaa • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Built an automated sensor-data cleansing pipeline (3.6 M data points) that cut model roll out from 3 days to less than 24 hours.
- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI

- Led a 4-engineer team to deliver ViCAS, enhancing model accuracy and reducing false positives by 33% through effective framing and edge-case resolution within 6 months.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Implemented real-time object tracking with YOLO-DeepSORT, maintaining consistent identities across frames and enabling reliable downstream analytics.
- Engineered synthetic training data pipelines using OpenCV and NumPy, reducing edge-case model development time from 6 days to 2 days for high-risk production environments.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.

EDUCATION

Master of Science in Robotics - 2:1

University of Bristol • Bristol, England

09/2023 - 09/2024

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

SKILLS

- **Languages & Paradigms:** C++ (14/17), object-oriented & functional programming, Python (3.x)
- **Data Streaming & Real-Time Processing:** Apache Kafka
- **Simulation & Modeling:** Jason (AgentSpeak), ROS & Gazebo
- **Version Control & CI/CD:** Git, GitHub Actions
- **Testing & Monitoring:** gtest, pytest